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Prenasalized stops and speech timing.[1]

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Prenasalized stops, that is, homorganic nasal+stop elements that behave as single phonological segments, raise a number of interesting questions concerning the relationship between phonological units and timing in speech. Do complex phonetic elements of this kind occupy the same duration as simpler elements, such as plain stops or nasals? Do prenasalized stops have the same timing pattern as a phonological sequence of nasal plus stop? How do prenasalized stops act with respect to rules which adjust the duration of neighboring segments? For example, would a vowel before a prenasalized stop be shortened by the widespread rule which shortens a vowel in a closed syllable (Maddieson 1985)? It has been argued that the status of prenasalized stops as single segments is directly related to their duration. They have been defined as nasal+stop sequences with the duration typical of other single segments (Herbert 1986). He, and Sagey (1986), in her dissertation on complex segments, both indicate that they would expect phonological consonant sequences to have longer durations than single segments regardless of whether the single segments are phonetically simple or complex. On the other hand, Ladefoged and Maddieson (1986) suggest that there is no demonstrated phonetic difference in timing between nasal+stop sequences and prenasalized stops. Purported *language-internal* contrasts between these elements actually involve a difference between geminate and single nasals before stops (as in Sinhalese), or between syllabic and non-syllabic nasals before stops. They suggest that deciding if a nasal+stop element is a prenasalized stop is not a phonetic issue but one which concerns solely the phonology of the language in question.

The Ladefoged and Maddieson view coincides with a principal point put forward by Browman and Goldstein (1986). At first glance, their data might appear to have answered many of the questions raised above. In the course of presenting a larger argument concerning phonological representations, they argue that prenasalized stops have similar temporal characteristics to nasal+stop sequences as well as to plain stops and that timing is not related to phonological status at a segmental level. Browman and Goldstein's demonstration of their point proceeds as follows. They monitored lower lip displacement in a set of real and nonsense words containing either a single medial labial consonant or a labial nasal+stop sequence, then superimposed lip movement traces from a selected token of each word from a single speaker of English, as in Figure 1. The coincidence of the lip traces in Figure 1 shows that "the overall envelope of the gestures is similar" and the consonantal portion has the same timing regardless of whether it is a single consonant or a consonant sequence. For data on prenasalized stops they used the Tanzanian Bantu language KiChaka (Chaga). Lip movement tracings from selected tokens of words beginning with simple labial segments were superimposed, together with one from a word beginning with /mb/, as shown in Figure 2. The data are from a speaker of the KiVunjo dialect. Again they point to the similarity between the various labial gestures and argue that plain stops and prenasalized stops in KiChaka are alike; specifically, they are produced with labial gestures that have the same time course. We now have essentially established a syllogism. English nasal+stop clusters are like (English) plain labial stops and nasals; KiChaka prenasalized stops are like (KiChaka) plain labial stops and nasals; therefore English stop+nasal clusters are like prenasalized stops.

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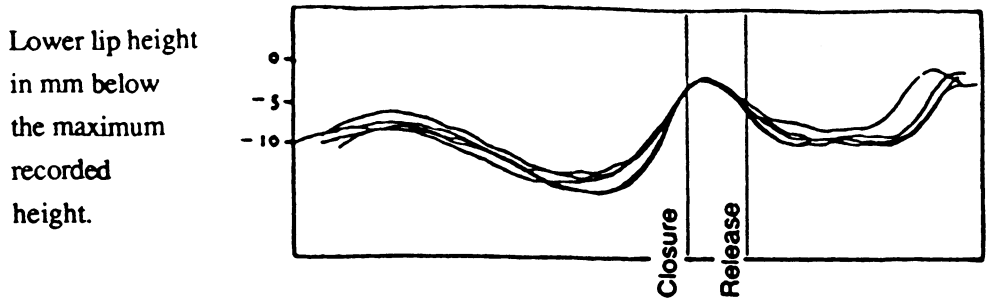

Figure 1. Superimposed lower lip movement trajectories of the English utterances *capper, cabber, cammer, camper* and *camber* (after Browman & Goldstein 1986). Trajectories are lined up at the peak of the labial articulatory gesture for each word. The value 0 is assigned to the highest lip position recorded in the session. Closure and release times indicated are for the utterance *capper*.

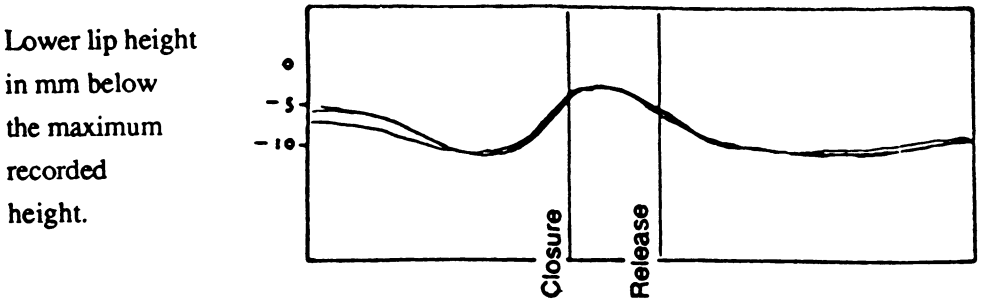

Figure 2. Superimposed lower lip movement trajectories of the KiChaka utterances /paka/ 'cat', /maka/ 'year' and /mbaka/ 'evil-doing' (after Browman & Goldstein 1986). Trajectories are lined up at the peak of the labial articulatory gesture for each word as in Figure 1. Closure and release times indicated are for the utterance /maka/.

Browman and Goldstein also suggest that there are additional temporal regularities in English which can be explained on the basis of a common gesture underlying both plain stops and nasals and the nasal + stop sequences. They point out that, using the same set of English words, Vatikiotis-Bateson (1984) found stressed vowel duration was the same before the single consonants /p/ and /b/ as before the matched nasal+stop cluster (i.e. /p/ and /mp/; /b/ and /mb/). Similarly, vowels were shorter before both /p/ and /mp/ than before /b/ and /mb/. Thus, vowel length was determined by stop voicing but not by status as single consonant or cluster. An expected shortening effect on the preceding vowel of the cluster (Lindblom and Rapp 1973, Klatt 1979) was not observed. As for KiChaka, Browman and Goldstein suggest that the phonological regularity of importance is that

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possible syllable onsets are those which consist of single oral articulatory gestures, whether or not nasality is involved.

Closer examination, however, suggests that there are some problems with Browman and Goldstein's approach to both English and KiChaka and these observations do not satisfactorily document a similarity between single stops and prenasalized ones.

Let us discuss the English case first. Pairs of words like "camper" and "capper" may be unsuited for testing effects of segmental organization, since there are good reasons for regarding the intervocalic consonant in words like "capper" as ambisyllabic (for example, /æ/ is a vowel that is normally limited to closed syllables). The vowel before /p/ is therefore in a syllable which contains a coda, just as the vowel in "camper" would be if the nasal + stop sequence has the articulatory organization of a cluster. The question of whether /mp/ is acting like a single consonant or a sequence cannot be answered by comparison with intervocalic /p/ since /p/ may be acting rather like a sequence itself, particularly with respect to effect on a preceding vowel.

The evidence from voicing-dependent vowel length adjustment rules is also unclear. If the similarity of application of these rules before plain stops and nasal + stop sequences is due to the fact that they share the same articulatory organization, that is, both are made with a single labial gesture with the same temporal characteristics, then we would expect this rule to be blocked from applying to the first vowel in pairs like the proper names 'Albert', 'Alpert', where a lingual gesture precedes the labial gesture. In fact, as is well known, in such cases there is not only a difference in the vowel duration, but also one in the duration of the lateral segments (Klatt 1979, Fourakis 1980). Such durational adjustments apply generally to liquids and nasals in English as well as to vowels (although Browman and Goldstein did not report this effect in the English materials they studied). Homorganicity is not a requirement for the rule to apply. Instead, only the voicing category of the last obstruent in the cluster is relevant. That the rule applies before nasal + stop sequences therefore does not indicate that they have single-segment-like temporal organization.

On the other side of the equation, KiChaka is not entirely convincing as a representative of a language which has a phonological category of prenasalized stops. The word-initial nasals in nasal + stop onsets are generally separate morphemes – prefixes specific to particular noun classes which may alternate with prefixes of CV or V structure. For example, the stem /-ku/ 'firewood' may be preceded by a singular prefix /u-/, or a 'mass' or collective prefix which takes the form of a nasal homorganic to the stem-initial consonant. These nasal prefixes are neither syllabic nor tone-bearing, but are simply incorporated in the syllable onset.[2] When preceded by a nasal, KiChaka stops become voiced (Nurse 1981: 142), hence the plural of 'firewood' is /ŋgu/. Otherwise - in absolute initial position or between syllabic segments - stops are always voiceless. The nasal prefix causes voicing assimilation of a following stop, by a rule which also governs morpheme-internal nasal + stop sequences, as in the noun /numba/ 'house'. There is nothing here to demonstrate the phonological unity of the nasal + stop elements. Their similarity in timing to initial plain stops is interesting, but does not clearly relate to the timing of phonologically unitary prenasalized stops, and is not known to have any relevance for the timing of adjoining segments, either at the phonetic level or in the operation of such phonological rules as compensatory lengthening.

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Therefore the questions concerning timing of prenasalized stops remain somewhat open. More extensive and less ambiguous phonetic data is required to address them. Data from Fijian has been collected with a view to assisting in this aim. Fijian is a language in which prenasalized stops are persuasively analyzed as single segments from the phonological point of view (Geraghty 1983). It has stops of two types, voiceless oral and voiced prenasalized. These two types of stops have the same distributional patterns. Unlike KiChaka, no boundaries fall between nasal and stop and a post-nasal voicing assimilation rule cannot be motivated by alternations. If we set aside the prenasalized stops for the moment, all syllables are open, and no consonant clusters occur. If they were regarded as a segment sequence, prenasalized stops in initial and medial positions would constitute the only clusters, and in medial position, their nasal portion would constitute the only syllable-closing consonants in the language. In short, there is nothing to suggest that prenasalized stops behave as other than single phonological segments.

Utterance-initial and word-medial prenasalized stops at bilabial, alveolar, post-alveolar and velar places of articulation were examined in the set of words shown in Table 1. The release of the postalveolar stops is sometimes trilled, so the Fijian orthographic convention of distinguishing them from the alveolar ones with an /r/ is adopted.[3]

TABLE 1. Fijian wordlist exemplifying prenasalized stops

	Initial	Prefixed	Medial	Initial & Medial
/mb/	mbaka	nambaka	kamba	mbamba
	"banyan tree"	(+ def. art)	"climb"	"slope"
/nd/	ndala	nandala	kanda	ndanda
	"dal"	(+ def. art)	"run"	"soft"
/ndr/	ndrata	mondrata	tandra	ndrandra
	"sit"	(+ 2 sg. pn.)	"dream"	*nonsense word*
/ŋg/	ŋgara	naŋgara	raŋga	ŋgaŋga
	"hole"	(+ def. art)	"show off"	"tough (of cooked fish)"

The word list was constructed to provide the opportunity to measure the acoustic duration of intervocalic prenasalized stops and to compare that with the duration of certain plain stops and liquids as well as to allow effects on preceding vowel duration to be measured. Plain nasals were not included because it was feared that nasalization of adjoining vowels would create problems in demarcating acoustic segments. In the event, the duration of the trill /r/ proved to be impossible to measure reliably, and nasals would have been a lesser problem. Only the vowel /a/ was used in the word list in order to eliminate any effect of intrinsic vowel duration variation.

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Prenasalized stops in both initial and medial positions of words were included to address another issue arising from the fact that the presence of a nasal portion in initial prenasalized stops is often not perceived by those who are not native speakers of Fijian or another similar language. This effect was hypothesized to be due to a difference in acoustic duration and/or amplitude between prenasalized stops in these two positions, rather than to a perceptual process alone. To refine this hypothesis, the word list also included medial prenasalized stops derived by adding a prefix to words with initial ones so as to examine if any residual effect of the initial/medial contrast remained, as it might if stress placement was the relevant controlling factor. Note that stress falls on the first syllable of the lexical entry and does not shift forward to the prefix. Hence the prefixed medial prenasalized stops are pre-stress, like the initial ones, whereas the underlyingly medial prenasalized stops in unprefixed forms are post-stress.

The word list was compiled and recorded with the generous assistance of Dr. Paul Geraghty. Recordings were made in a sound-attenuated room belonging to the Fiji Broadcasting Service in the Government buildings in Suva, Fiji. Eleven speakers representing three main varieties of Fijian were recorded, with the majority (six) being speakers of what is considered the standard dialect, the others from Lau (two) and Kandavu and Kandavu (orthographically, Kadavu) dialects. The subjects were all in the 30-50 age range. Seven were female - three of the Standard Fijian speakers, two of the Kandavu and one of the Lau speakers. Each read the list twice, once across row by row and once down column by column.

Duration measurements were made on wide-band spectrograms of all the words spoken by the eleven subjects. A sample spectrogram is shown in Figure 3. Acoustic duration of the prenasalized stops was measured from onset of nasal murmur to stop release (the stop component is usually very brief and not separately measurable). Durations of the other medial consonants whose acoustic boundaries could be determined were also measured, as were the durations of the initial nasals in the prefixed forms. In addition, the duration of the majority of the stressed vowels was measured from preceding consonant release to following consonant closure.

The resulting measurements were analyzed using an analysis of variance procedure. Means were compared using the Scheffé F-test. No significant interaction was found between the dialect groups and place of articulation in the prenasalized stops, ($F(6,414) = 0.876$, $p = .51$). This means that the pattern of duration variations in prenasalized stops is similar across the dialects. However, the dialects differ in the realization of some of the other consonants (for example, voiceless velar stops are replaced by velar fricatives in Kandavu). Hence some results are given pooled across all speakers, while others include only the Standard Fijian speakers. Durations of the medial prenasalized stops are given in Table 2, together with the durations of the three other measurable intervocalic consonants, voiceless stops /t/ and /k/ and the lateral /l/. No significant difference was found between underlyingly medial prenasalized stops and those which are medial through prefixation ($F(1,247) = 0.477$, $p = .49$), so measurements on these two categories have been pooled. Bilabial and alveolar prenasalized stops are significantly longer than postalveolar and velar ones, that is, for each pairing of bilabial or alveolar with postalveolar or velar there is a significant difference (lowest value of $F(5,251)$ is 4.082, $p<.01$). But the durations of these two groups of prenasalized stops are very comparable to the durations of the two plain stops in the data, which also reflect the difference between alveolar and velar place of

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articulation. The alveolar lateral continuant falls within the same range of durations covered by the prenasalized stops. Hence we may conclude that the acoustic duration patterns of the prenasalized stops are very comparable to those of simple segments in Fijian.

TABLE 2. Mean duration of medial consonants

	mean	s.d.	n
All speakers			
/mb /	132	22.2	62
/nd /	131	23.8	64
/ndr /	114	25.5	65
/ŋg /	114	32.6	65
Standard Fijian speakers only			
/t /	125	19.5	24
/k /	116	30.1	24
/l /	117	12.3	21

Our measurements also indicate that vowel length before prenasalized stops is comparable to that before other consonants. Results are given in Table 3. Overall, vowel duration remains consistently close to 140 ms in the various contexts measured. We may examine these data for effects of consonant voicing and prenasalization, place of articulation, and, to a limited degree, manner. Voicing and prenasalization are of course confounded, but vowel duration seems unaffected by whether a voiceless stop or voiced prenasalized stop follows. Mean duration of the /a/ vowel before the voiceless stop /t/ for the Standard Fijian speakers is 142 ms, while before the prenasalized stop with the same place of articulation it is 139 ms. Needless to say, these durations are not significantly different ($F(1,44) = 0.234$, $p = .63$). Nor is there evidence to support a tradeoff between place-dependent consonant duration (Table 2) and preceding vowel duration. Before /k/ the vowel duration of 130 ms is actually a little shorter than before /t/, even though /k/ is itself shorter than /t/. Although the vowel before /ŋg/ in /ŋgaŋga/ is a little longer than the overall mean at 148 ms, there is no general evidence that the voicing in the prenasalized stops conditions a longer preceding vowel, as the alveolar examples demonstrate. Further, before both /mb/ in /kamba/ and /ndr/ in /tandra/ mean vowel duration is 141 ms. The 18 ms difference before /k/ vs /ŋg/ therefore seems aberrant. It is possible that the shorter vowel before /k/ results accidentally from the fact that the tokens containing medial /k/ have a bilabial stop *before* the vowel. Shorter vowels occur after bilabials than alveolars in Danish (Fischer-Jørgensen 1979) and perhaps in English (Halle and Stevens 1967) and other languages where bilabial closures are longer. Fijian did not show any effect of following consonant manner in the one case tested by our data: mean vowel duration before intervocalic /l/ was measured at 142 ms.

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TABLE 3. Mean stressed vowel durations before consonants

Standard Fijian speakers only

environment			mean	s.d	n
alveolar					
	/_t/	(ndrata, mondrata)	142	28.0	24
	/_nd/	(kanda, ndanda)	139	12.2	22
velar					
	/_k/	(mbaka, nambaka)	130	22.2	24
	/_ŋg/	(ŋgaŋga)	148	17.4	12
bilabial					
	/_mb/	(kamba)	141	17.1	11
post-alveolar					
	/_ndr/	(tandra)	141	15.9	12
lateral					
	/_l/	(ndala)	142	12.4	12

In these results we see evidence that Fijian vowels can be considered as isolated from the effects of following consonants, a view which is consistent with the idea that all the consonants we have examined, including the prenasalized stops, are single segments that belong exclusively to the syllable of which they form the onset. However, another possibility must be considered. If vowels were subject to a shortening effect when a cluster follows but also subject to a lengthening effect when a voiced segment follows, these two effects might cancel each other to produce comparable length before voiced nasal + stop clusters and voiceless single stops. These effects are usually regarded as (at least in part) compensatory adjustments, associated with the longer duration of clusters (producing shortening) and the shorter duration of voiced consonant closures (producing lengthening) (Fischer-Jørgensen 1964, Lehiste 1970, Lindblom and Rapp 1973). We note the absence in Fijian of the other type of widely reported compensatory adjustment, dependent on place of articulation of the following consonant. In view of this, we prefer to adopt the position that vowel duration is unaffected by following consonants, rather than to assume that they are affected by two separate but separately unobservable compensatory duration adjustments.[4]

As for the question of an initial/medial difference in the prenasalized stops, the initial ones do have a significantly shorter acoustic duration than the medial ones, the difference being 39 ms, ($F(1,418) = 10.173$, $p < .0001$). The means are 95 ms for bilabial, 85 ms for alveolar, 83 ms for postalveolar and 72 ms for velar. The velar duration is significantly shorter than the bilabial ($p < .01$) and alveolar ($p < .05$) durations. Initial prenasalized stops also have noticeably lower amplitude, as exemplified in the amplitude envelope on the spectrogram shown in Figure 3. As noted above, when underlying initial prenasalized stops are preceded by an unstressed prefix, they become indistinguishable in duration from underlyingly medial ones. Therefore it is clearly the difference in utterance

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position (initial vs medial) which controls the duration difference, not the position of the prenasalized stop in relation to stress (pre-stress vs post-stress). Parenthetically we might note that there is an indication that consonants initiating an unstressed syllable are shortened in Fijian: the initial nasals in unstressed prefix syllables are significantly shorter than initial prenasalized stops (/n/ = 45 ms, /m/ = 51 ms). The short acoustic duration of utterance-initial prenasalized stops is possibly simply due to a delay in the onset of vocal cord vibration from a pre-speech setting, and may not reflect a briefer articulatory gesture. However, this effect surely contributes to the lesser salience of the nasal portion of prenasalized stops in initial position.

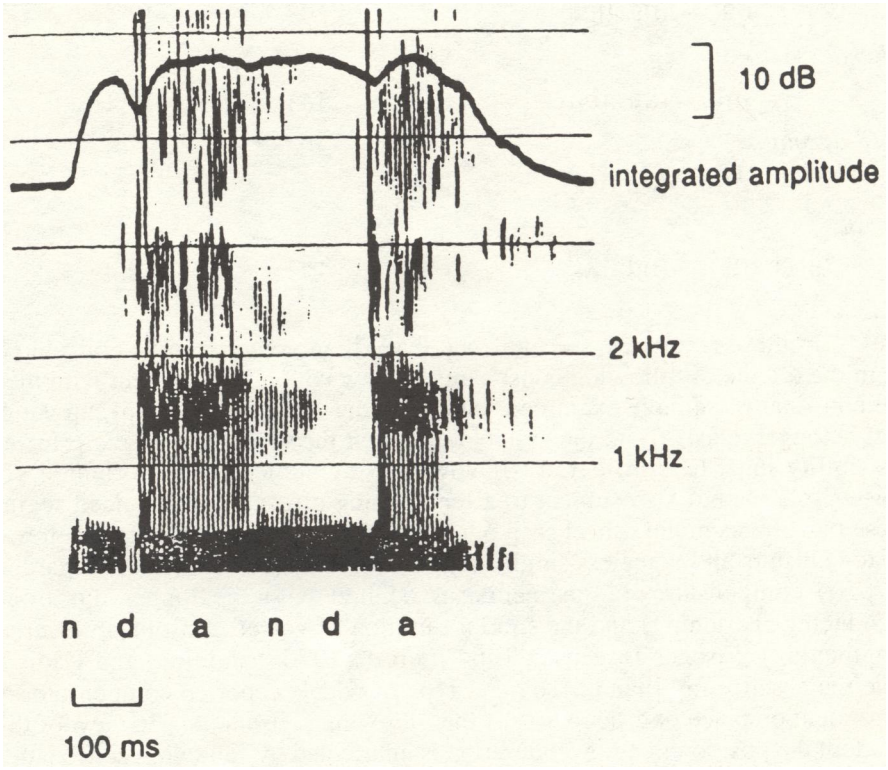

Figure 3. Broad-band spectrogram of the Fijian word /ndanda/ 'soft' spoken by a speaker of Standard Fijian. Duration and amplitude of initial and medial prenasalized stops can be compared on this figure.

In conclusion, these results indicate that the role of prenasalized stops in the timing pattern of Fijian is generally similar to that of other single segments. They neither have

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longer duration themselves nor do they shorten a preceding vowel, as might be expected if they had a timing pattern like that of geminate consonants or consonant clusters. Prenasalized stops, at least in Fijian, therefore do possess the timing characteristics stated by some linguists to be part of their definition. We have also shown that Fijian temporal patterns appear to be strictly based on the CV syllable, place and manner differences in following consonants having no effect on preceding vowel duration.

Comparison with appropriate data from languages in which the phonological structure of nasal+stop elements is unambiguously that of a sequence is required before conclusions can be drawn about whether, in general, timing patterns distinguish between prenasalized stops and nasal+stop clusters or demonstrate that they are alike at the level of production.

Acknowledgments

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Notes

1. This is an expanded version of a paper presented at the 115th Meeting of the Acoustical Society of America, Seattle, Washington, May 16-20, 1988.

2. On the other hand, syllabic nasals arise in KiChaka from contraction of an NV-syllable, and appear in a few loanwords, such as /mpaka/ 'boundary' (from Swahili).

3. Fijian orthography differs from our transcription in that the nasal portion of prenasalized stops is not represented; symbols that represent simple voiced stops are used except that /ŋg/ is written 'q'.

4. Unfortunately, a combination of the limitations of our data set and the restrictions on Fijian segments impede further investigation of these questions. If medial nasals, non-prenasalized voiced stops, and clusters of different types could have been examined, any effects of voicing and cluster structure could have been separated. Nasals might have been included in the word list, but the others are impermissible in Fijian.

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